

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import CountVectorizer

from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import accuracy_score
```

```
# Load Dataset
```

```
data = pd.read_csv("spam.csv")
```

```
# Convert labels
```

```
data['v1'] = data['v1'].map({'ham': 0, 'spam': 1})
```

```
# Features and Labels
```

```
X = data['v2']
```

```
y = data['v1']
```

```
# Split Data
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Convert text into numbers
```

```
vectorizer = CountVectorizer()
```

```
X_train_vec = vectorizer.fit_transform(X_train)
```

```
X_test_vec = vectorizer.transform(X_test)
```

```
# Train Model
```

```
model = MultinomialNB()
```

```
model.fit(X_train_vec, y_train)
```

```
# Predict
```

```
predictions = model.predict(X_test_vec)
```

```
# Accuracy
```

```
accuracy = accuracy_score(y_test, predictions)
```

```
print("Accuracy:", accuracy)
```

```
# Test Message
```

```
message = ["Congratulations! You won a free gift."]
```

```
message_vec = vectorizer.transform(message)
```

```
result = model.predict(message_vec)
```

```
if result[0] == 1:
```

```
    print("Spam Email")
```

```
else:
```

```
    print("Not Spam Email")
```